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PyDTS: A Python Package for Discrete-Time Survival Analysis with Competing Risks and Optional Penalization

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Summary

Time-to-event (survival) analysis models the time until a pre-specified event occurs. When time is measured in discrete units or rounded into intervals, standard continuous-time models can yield biased estimators. In addition, the event of interest may belong to one of several mutually exclusive types, referred to as competing risks, where the occurrence of one event prevents the occurrence or observation of the others. *PyDTS* is an open-source Python package for analyzing discrete-time survival data with competing-risks. It provides regularized estimation methods, model evaluation metrics, variable screening tools, and a simulation module to support research and development.

Statement of need

Time-to-event analysis is applied when the outcome of interest is the time until a pre-specified event occurs. In some settings, the time variable is inherently or effectively discrete, for example, when time is measured in weeks or months, or when event times are rounded or grouped into intervals. Competing risks arise when observations are at risk of experiencing multiple mutually exclusive event types, such that the occurrence of one event precludes the occurrence or observation of the others. Discrete-time survival data with competing risks are encountered across a wide range of scientific disciplines. For instance, in healthcare, the time to death from cancer is often recorded in months, with death from other causes considered a competing event.

While excellent Python packages for continuous-time survival-analysis exist (Davidson-Pilon, 2019; Pölsterl, 2020), a comprehensive, user-friendly Python toolkit specifically designed for discrete-time survival analysis is still missing. Moreover, in the continuous-time setting, competing-risks data can often be analyzed using methods developed for non-competing events, since the full likelihood function factorizes into separate likelihoods for each cause-specific hazard function (Kalbfleisch & Prentice, 2011). In contrast, this factorization does not hold in the discrete-time setting (Lee et al., 2018; Meir & Gorfine, 2025), and dedicated estimation procedures are required to correctly account for the competing risk structure.

PyDTS bridges this gap by providing tools for analyzing discrete-time survival data with competing risks, designed to support both expert and non-expert researchers. Specifically, it offers:

- Discrete-time competing-risks regression models, based on the methods of Lee et al. (2018) and Meir & Gorfine (2025).
- Automated procedures for hyperparameter tuning.
- Sure Independence Screening methods for feature selection (Zhao & Li, 2012).

- Model evaluation metrics for predictive accuracy and calibration (Meir & Gorfine, 2025).
- Simulation tools for generating synthetic datasets for research and testing.

To the best of our knowledge, *PyDTS* is the first open-source Python package dedicated for discrete-time survival analysis with competing risks. Details on the statistical models and methods implemented in *PyDTS* are summarized in the package documentation and described in great detail in Meir & Gorfine (2025).

Package Details

Dependencies

PyDTS can be easily installed via *PyPI* as follows:

```
pip install pydts
```

This command automatically installs the following dependencies:

- NumPy (Harris et al., 2020) (<2.0.0)
- pandas (McKinney, 2010)
- lifelines (Davidson-Pilon, 2019)
- SciPy (Virtanen et al., 2020)
- scikit-learn (Pedregosa et al., 2011)
- statsmodels (Seabold & Perktold, 2010)

Key Features

1. **Estimation procedures:** Two methods are implemented, `TwoStagesFitter` of Meir & Gorfine (2025), and `DataExpansionFitter` of Lee et al. (2018). The `TwoStagesFitter` supports both regularization and the inclusion of time-dependent covariates, features that are not available in the `DataExpansionFitter` implementation.
2. **Sure Independence Screening:** The `SISTwoStagesFitter` class implements the Sure Independence Screening (SIS) of Zhao & Li (2012). SIS is a powerful dimensionality reduction technique designed for ultra-high-dimensional settings, where the number of covariates far exceeds the number of observations, a situation often encountered in genomic studies and other high-throughput domains. It works by filtering out a large number of uninformative covariates based on their marginal association with the outcome. After screening, penalized variable selection methods (e.g., LASSO) are typically applied to the reduced set of covariates to perform more refined modeling and selection.
3. **Evaluation Metrics:** The package includes functions for computing key performance metrics for discrete-time survival data with competing risks and right-censoring, including the cause-specific cumulative/dynamic area under the receiver operating characteristic curve (AUC) and the Brier score (BS). Formal definitions of all implemented evaluation metrics are provided in Meir & Gorfine (2025).
4. **Hyperparameters tuning:** The package provides automated procedures for hyperparameter selection, including grid search combined with cross-validation, enabling robust model calibration and improved generalization performance.
5. **Data Generation:** The `EventTimesSampler` module facilitates the generation of discrete-time survival data with competing risks and right censoring. Given user-specified model parameters, including the number of discrete event times, true regression coefficients, and covariate values for each observation, `EventTimesSampler` simulates both event times and event types. The module supports two types of right censoring: administrative censoring, applied when the simulated event time exceeds a user-defined maximum follow-up duration, and random censoring, which can be either covariate-dependent or independent. This flexible simulation framework is useful for benchmarking models, testing estimation procedures, and conducting methodological research.

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Case Study

The utility of *PyDTS* is demonstrated through an analysis of patients' length of stay (LOS) in intensive care unit (ICU), conducted by Meir & Gorfine (2025). This analysis uses the publicly accessible, large-scale Medical Information Mart for Intensive Care (MIMIC-IV, version 2.0) dataset (Goldberger et al., 2000; Johnson et al., 2022).

Meir & Gorfine (2025) developed a discrete-time survival model to predict ICU LOS based on patients' clinical characteristics at admission. The data comprises 25,170 ICU admissions, with LOS recorded in discrete units from 1 to 28 days, resulting in many patients sharing the same event time on each day. Three competing events are considered: discharge to home (69.0%), transfer to another medical facility (21.4%), and in-hospital death (6.1%). Patients who left the ICU against medical advice (1.0%) are treated as right-censored, and administrative censoring is applied to those hospitalized for more than 28 days (2.5%). The analysis includes 36 covariates per patient. For a full description of the data, see Meir & Gorfine (2025).

Three estimation procedures were compared: the method of Lee et al. (2018) without regularization, two-step approach of Meir & Gorfine (2025) without and with LASSO regularization. LASSO was implemented using a grid search with 4-fold cross-validation to select the optimal hyperparameters by maximizing the global-AUC metric. Detailed results of the case study are presented in Meir & Gorfine (2025).

The code below illustrates a cross-validated grid search that selects the optimal LASSO penalty parameters for each competing event.

```
import numpy as np
from pydts.cross_validation import PenaltyGridSearchCV

penalizers = np.exp(range(-12, -1))
penalty_cv_search = PenaltyGridSearchCV()
gauc_cv_results = penalty_cv_search.cross_validate(
    full_df=mimic_df, l1_ratio=1, penalizers=penalizers, n_splits=4)
```

where `mimic_df` is the full dataframe containing the covariates, an event-type column, an event-time column, and an event indicator column; `penalizers` is the set of penalization values evaluated for each risk, denoted as η_j, with $j = 1, 2, 3$; `n_splits` is the number of folds; and `l1_ratio` controls the balance between L1 and L2 regularization, with `l1_ratio` = 1 corresponding to pure L1 (LASSO) regularization. The results are shown in Figure 1.

Additional examples demonstrating *PyDTS*'s functionality are also provided in Meir & Gorfine (2025) and in the package documentation. These include analyses with regularized regression across varying sample sizes and levels of covariates' correlation, as well as the application of Sure Independence Screening in ultra-high-dimensional settings (Zhao & Li, 2012). These examples make use of the package's built-in data generation tools, underscoring its usefulness for methodological development and evaluation.

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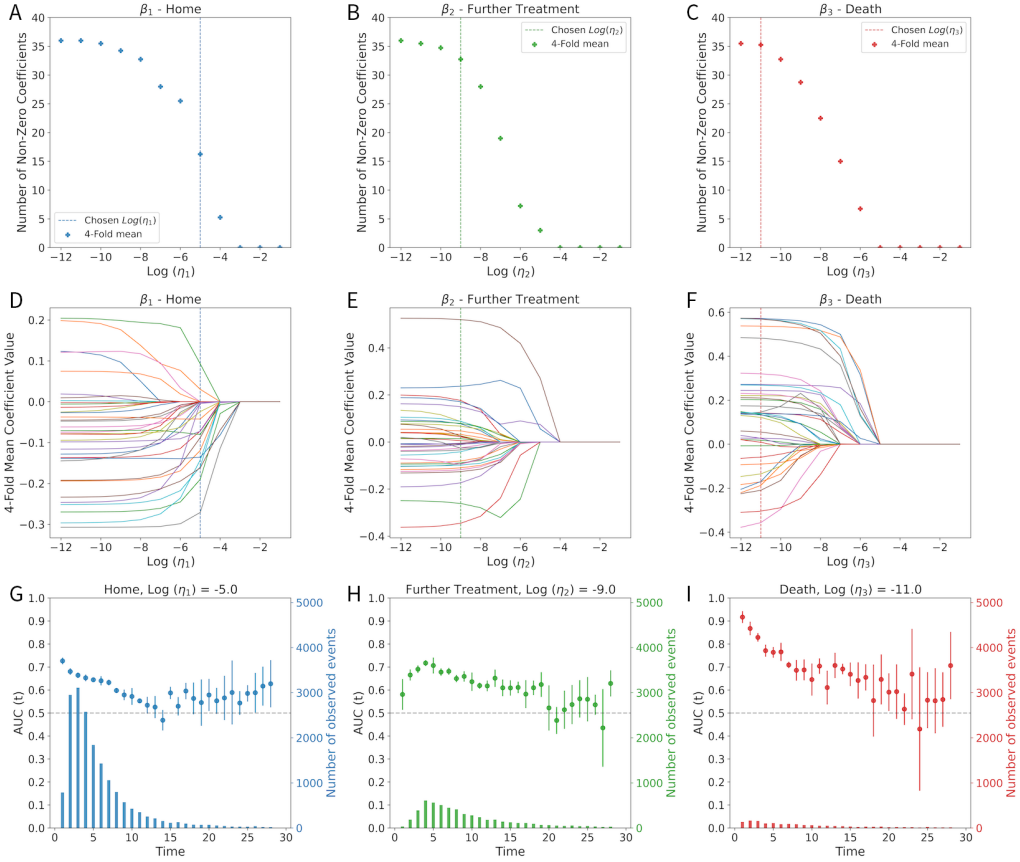

Figure 1: MIMIC dataset - LOS analysis. Regularized regression with 4-fold CV. The selected values of η_j are shown in dashed-dotted lines on panels **A-F**. **A-C.** Number of non-zero coefficients for $j = 1, 2, 3$. **D-F.** The estimated coefficients, as a function of η_j, $j = 1, 2, 3$. **G-I.** Mean (and SD bars) of the 4 folds $\widehat{\text{AUC}}_j(t)$, $j = 1, 2, 3$, for the selected values $\log \eta_1 = -5$, $\log \eta_2 = -9$ and $\log \eta_3 = -11$. The number of observed events of each type is shown by bars.

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References

Davidson-Pilon, C. (2019). Lifelines: Survival analysis in python. *Journal of Open Source Software*, *4*(40), 1317. https://doi.org/10.21105/joss.01317

Goldberger, A. L., Amaral, L. A. N., Glass, L., Hausdorff, J. M., Ivanov, P. Ch., Mark, R. G., Mietus, J. E., Moody, G. B., Peng, C.-K., & Stanley, H. E. (2000). PhysioBank, PhysioToolkit, and PhysioNet: Components of a new research resource for complex physiologic signals. *Circulation*, *101*(23). https://doi.org/10.1161/01.CIR.101.23.e215

Harris, C. R., Millman, K. J., van der Walt, S. J., Gommers, R., Virtanen, P., Cournapeau, D., Wieser, E., Taylor, J., Berg, S., Smith, N. J., Kern, R., Picus, M., Hoyer, S., Kerkwijk, M. H. van, Brett, M., Haldane, A., Río, J. F. del, Wiebe, M., Peterson, P., … Oliphant, T. E. (2020). Array programming with NumPy. *Nature*, *585*, 357–362. https://doi.org/10.

1038/s41586-020-2649-2

Johnson, A., Bulgarelli, L., Pollard, T., Horng, S., Celi, L. A., & Mark, R. (2022). MIMIC-IV (version 2.0). *PhysioNet*. https://doi.org/10.13026/7vcr-e114

Kalbfleisch, J. D., & Prentice, R. L. (2011). *The statistical analysis of failure time data* (2nd ed.). John Wiley & Sons. https://doi.org/10.1002/9781118032985

Lee, M., Feuer, E. J., & Fine, J. P. (2018). On the analysis of discrete time competing risks data. *Biometrics*, *74*(4), 1468–1481. https://doi.org/10.1111/biom.12881

McKinney, W. (2010). Data structures for statistical computing in python. *Proceedings of the 9th Python in Science Conference*, 51–56. https://doi.org/10.25080/Majora-92bf1922-00a

Meir, T., & Gorfine, M. (2025). Discrete-Time Competing-Risks Regression with or without Penalization. *Biometrics*, *81*. https://doi.org/10.1093/biomtc/ujaf040

Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., Vanderplas, J., Passos, A., Cournapeau, D., Brucher, M., Perrot, M., & Duchesnay, E. (2011). Scikit-learn: Machine learning in python. *Journal of Machine Learning Research*, *12*, 2825–2830. https://doi.org/10.5555/1953048.2078195

Pölsterl, S. (2020). Scikit-survival: A library for time-to-event analysis built on top of scikit-learn. *Journal of Machine Learning Research*, *21*(212), 1–6. http://jmlr.org/papers/v21/20-729.html

Seabold, S., & Perktold, J. (2010). *Statsmodels: Econometric and statistical modeling with python*. https://doi.org/10.25080/Majora-92bf1922-011

Virtanen, P., Gommers, R., Oliphant, T. E., Haberland, M., Reddy, T., Cournapeau, D., Burovski, E., Peterson, P., Weckesser, W., Bright, J., van der Walt, S. J., Brett, M., Wilson, J., Millman, K. J., Mayorov, N., Nelson, A. R. J., Jones, E., Kern, R., Larson, E., … Mulbregt, P. van. (2020). SciPy 1.0: Fundamental algorithms for scientific computing in python. *Nature Methods*, *17*, 261–272. https://doi.org/10.1038/s41592-019-0686-2

Zhao, S. D., & Li, Y. (2012). Principled sure independence screening for cox models with ultra-high-dimensional covariates. *Journal of Multivariate Analysis*, *105*(1), 397–411. https://doi.org/10.1016/j.jmva.2011.08.002

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